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Thesis Group ID: T26101243

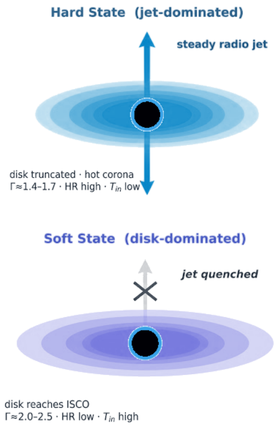
Latent Regime Discovery in Black Hole Accretion Disks via Structural Invariance

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SCAN FOR THE FULL THESIS

Abstract

Black Hole Accretion systems shift between physically distinct states, but observational data arrives unlabelled and existing causal-discovery methods require predefined environment labels or known regime counts. We propose LIRA, a causal AI framework that recursively partitions unlabelled time series using a lagged Chow-test invariance criterion gated by BIC, then recovers regime-specific causal graphs with DYNOTEARS - discovering hidden accretion regimes and their causal structure with no prior labels or state count.



Research Objectives

- Model accretion observables in log-linear structural causal framework derived from the Fundamental Plane of Black Hole Activity
- Develop a Chow-test invariance criterion, extended with a lagged AR(1) term, to detect instability in causal mechanisms
- Build a greedy recursive binary-partitioning algorithm that locates latent regime boundaries automatically
- Recover regime-specific causal graphs with DYNOTEARS and validate effects via refutation, sensitivity analysis, and graph falsification

Key Results at a Glance

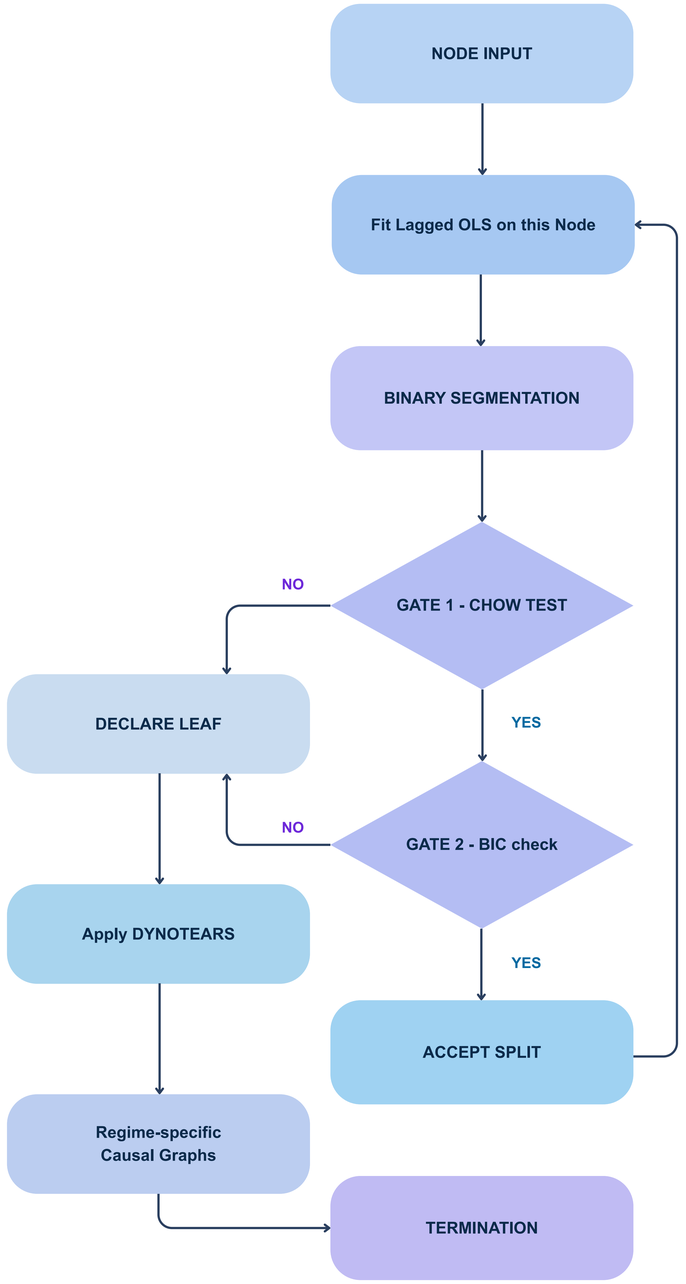
0.998
Adjusted Rand Index for regime recovery (best baseline: 0.587)

0.04%
Boundary localisation error (4 out of 10,000 observations)

≤ 0.017
Max. error on any recovered structural coefficient

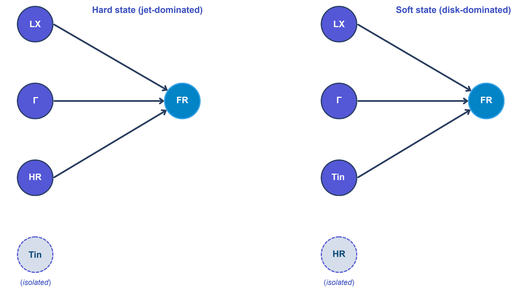
K = 2
Latent Regime found automatically - no count supplied in advance

Methodology & Workflow

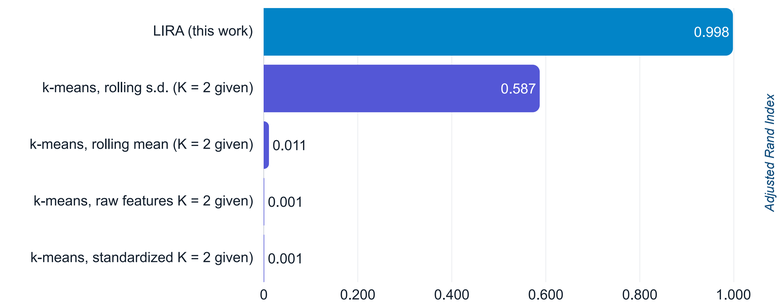


Regime-Specific Causal Structures

DYNOTEARS recovers each regime's causal graph independently. Every edge is correct and the isolated variable is correctly excluded in each state - the hard state is jet-dominated, the soft state is disk-dominated.



Regime Recovery vs. Baselines



Causal Effects

Hard state (jet-dominated)

cause ↓ effect →	LX	Γ	T_{in}	HR	FR
LX	+0.851†	0	0	0	+0.711
Γ	0	+0.854†	0	0	−0.296
T_{in}	0	0	+0.843†	0	*isolated*
HR	0	0	0	+0.854†	−0.398
FR	0	0	0	0	+0.498

Soft state (disk-dominated)

cause ↓ effect →	LX	Γ	T_{in}	HR	FR
LX	+0.851†	0	0	0	+0.180
Γ	0	+0.839†	0	0	+0.199
T_{in}	0	0	+0.838†	0	−0.383
HR	0	0	0	+0.847†	*isolated*
FR	0	0	0	0	+0.506

Row = cause, column = effect. Blue = positive, violet = negative; deeper shade = larger magnitude. 0 = no edge in the recovered graph.
† = lag-1 self-effect (DYNOTEARS weight); all other entries are backdoor-adjusted effect estimates.